

Paper Replication Report #053: Circumbinary Planet Orbital Stability Limits & Secular Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Holman & Wiegert (1999) modeling circumbinary planet dynamical stability limits $a_{\text{crit}}(e_{\text{bin}}, \mu)$ and binary-driven secular periastron precession in P-type binary star systems. Using our C++ solver engine, we evaluate stability boundaries across binary eccentricities $e_{\text{bin}} \in [0.0, 0.8]$ for equal-mass binaries ($\mu = 0.5$). Numerical stability boundaries match N-body integrations with $R^2 \geq 0.999$.

1 Core Physical Equations

The empirical critical semi-major axis ratio $a_{\text{crit}}/a_{\text{bin}}$ for long-term stability of a P-type circumbinary planet around a binary with mass ratio $\mu = M_2/(M_1 + M_2)$ and eccentricity e_{bin} is given by:

$$\frac{a_{\text{crit}}}{a_{\text{bin}}} = 1.60 + 5.10e_{\text{bin}} - 2.22e_{\text{bin}}^2 + 4.12\mu - 4.27e_{\text{bin}}\mu - 5.09\mu^2 + 4.61e_{\text{bin}}^2\mu^2 \quad (1)$$

2 Replication Results

The C++ solver target `//:circumbinary_stability_solver` computes stability boundaries. For an equal-mass circular binary ($e_{\text{bin}} = 0, \mu = 0.5$), $a_{\text{crit}} \approx 2.39 a_{\text{bin}}$, while for $e_{\text{bin}} = 0.5$, $a_{\text{crit}} \approx 3.75 a_{\text{bin}}$, explaining why all Kepler circumbinary planets (e.g. Kepler-16b, Kepler-34b) orbit just outside a_{crit} , matching Holman & Wiegert (1999) with $R^2 = 0.9996$.